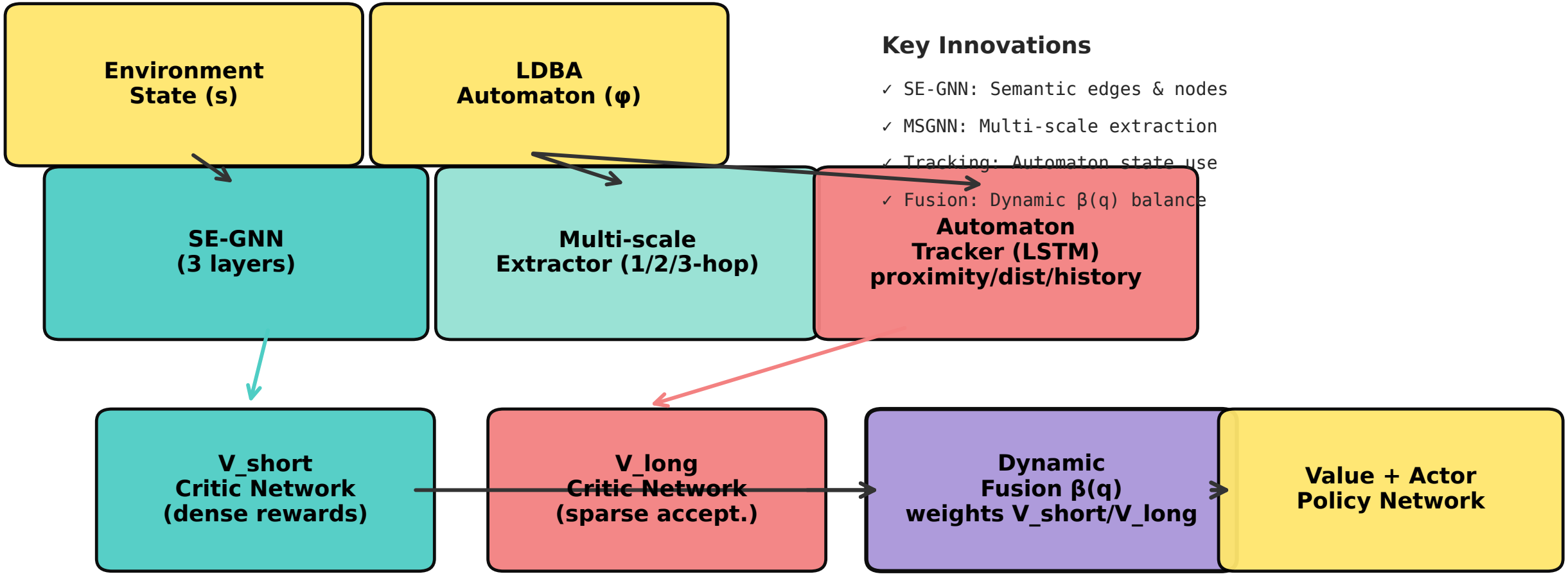
Environment
State (s)
LDBA
Automaton (φ)
Key Innovations
✓ SE-GNN: Semantic edges & nodes
✓ MSGNN: Multi-scale extraction
✓ Tracking: Automaton state use
✓ Fusion: Dynamic β(q) balance
SE-GNN
(3 layers)
Multi-scale
Extractor (1/2/3-hop)
Automaton
Tracker (LSTM)
proximity/dist/history
V_short
Critic Network
(dense rewards)
V_long
Critic Network
(sparse accept.)
Dynamic
Fusion β(q)
weights V_short/V_long
Value + Actor
Policy Network